

Principles of Error Management

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Traditional Error Management

Selection

Training...and
Retraining

Work planning

Job Cards

Tags,
Reminders

Shift
handovers

License-to-
work

Certifications

Audits

Sign-Offs

Procedures,
Manuals,
Rules

Discipline
Procedures

Human Error is Both Universal and Inevitable

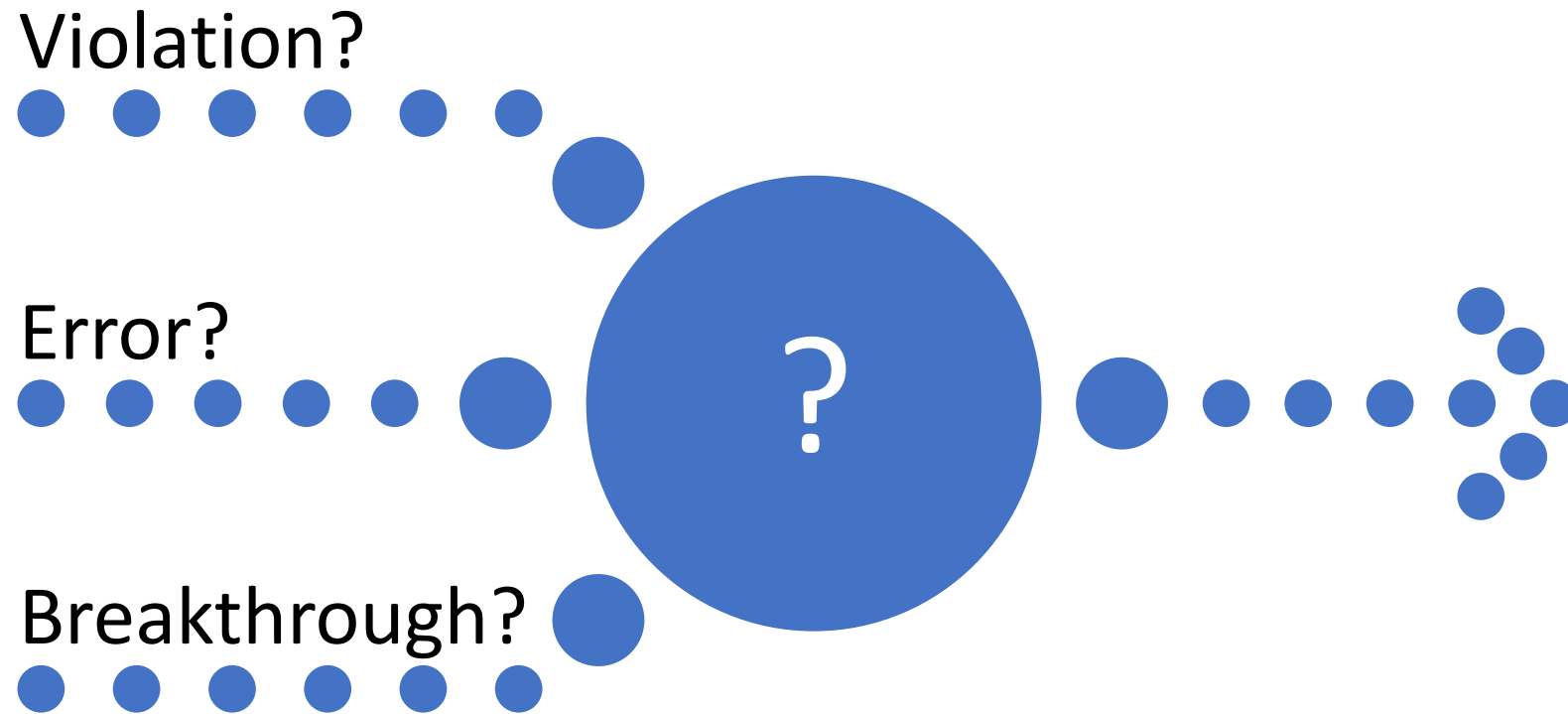
Errors are accidental and are not malicious.

Even violations are rarely malicious...they are actually often well intentioned.

Error cannot be eliminated...nor should it be.



Errors are not Intrinsically Bad



You Cannot Change the Human Condition, but you can Change the Conditions in which People Work

Equipment Failures

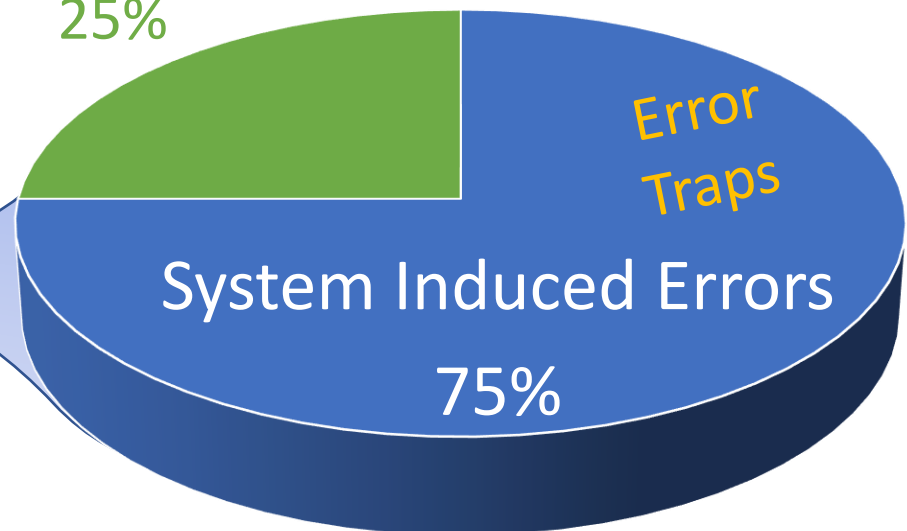
10%



Operational Upsets

Slips, Trips, Lapses

25%



Human Errors

The Best People can make the Worst Mistakes

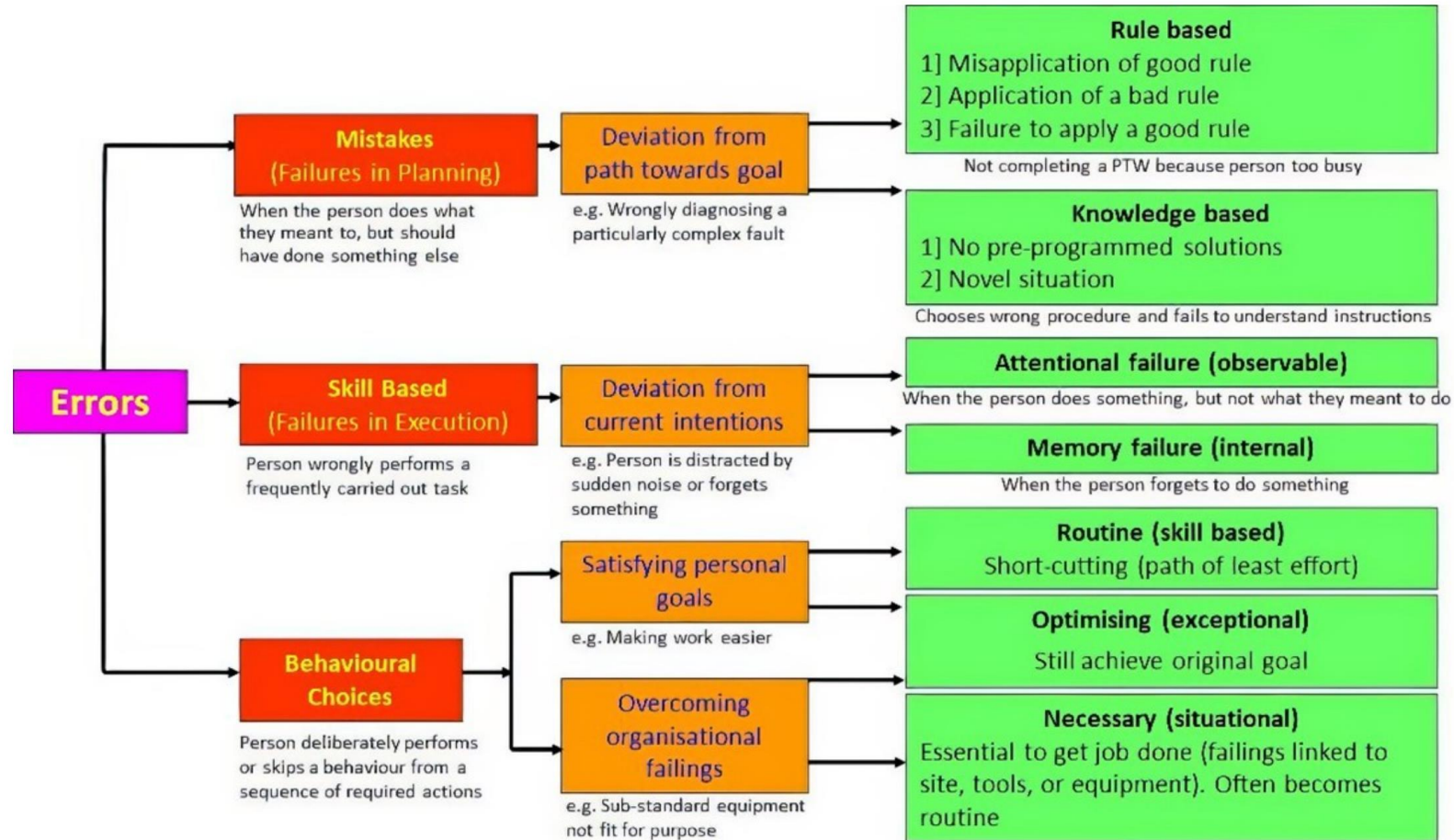
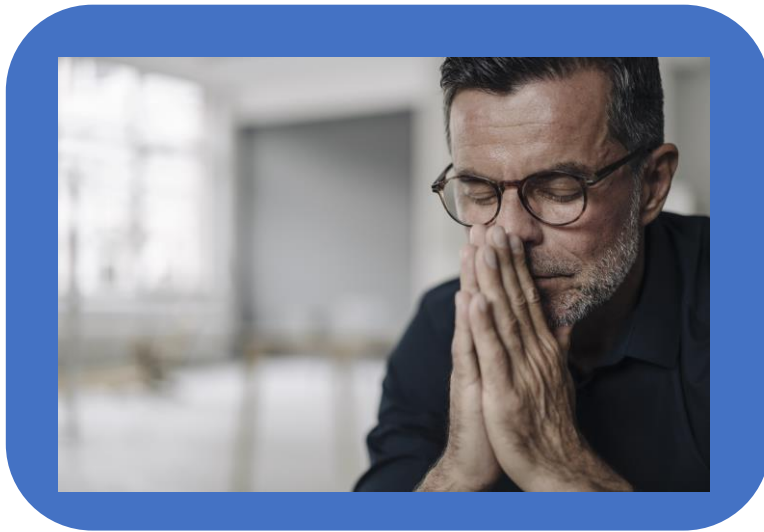


Figure 1: Reason's (1990) Human Error Model

People Cannot Easily Avoid those Actions they did not Intend to Commit

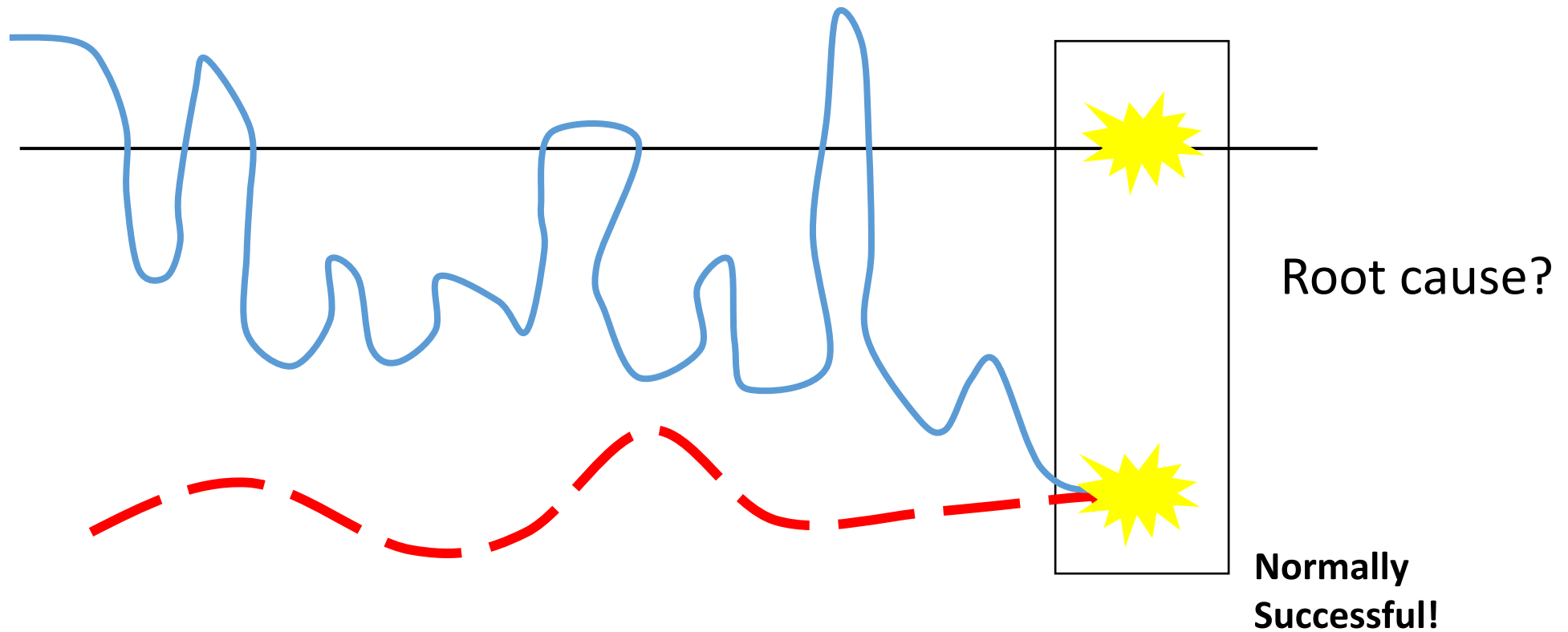


Blame vs.

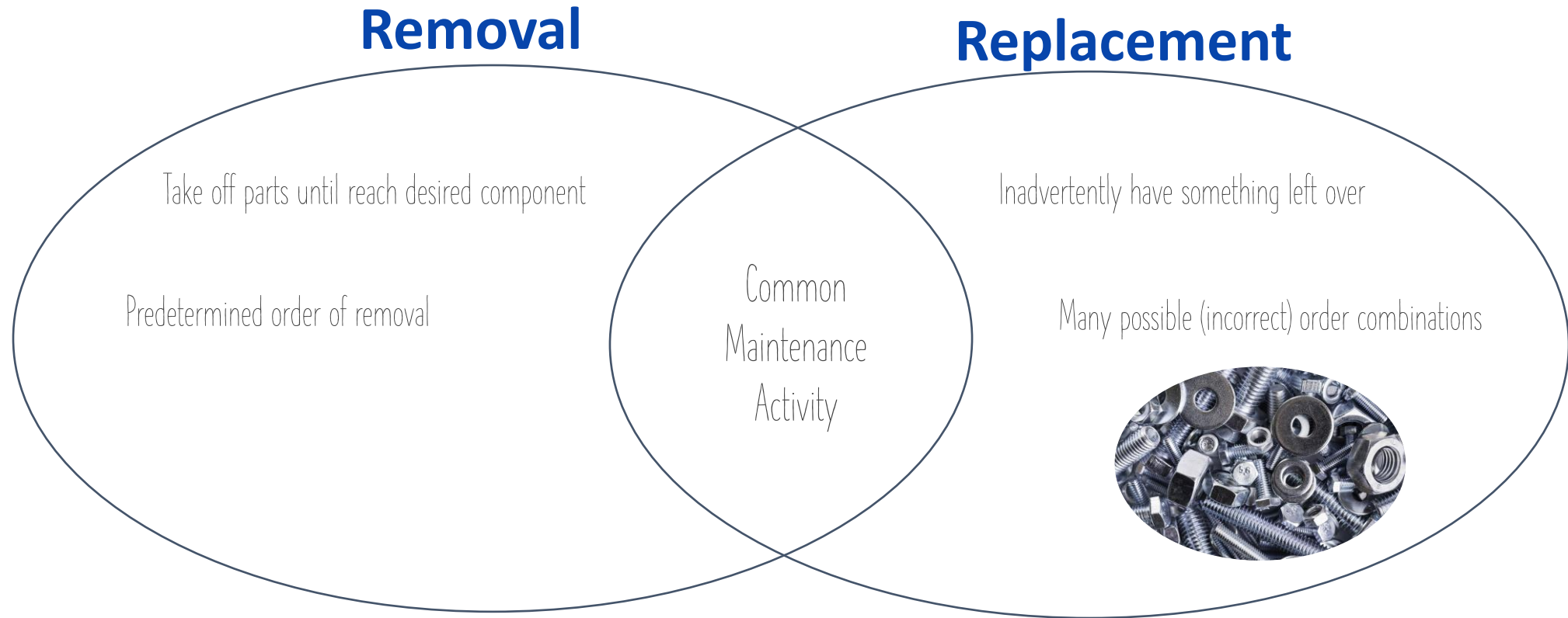


Accountability

Errors are Consequences rather than Causes

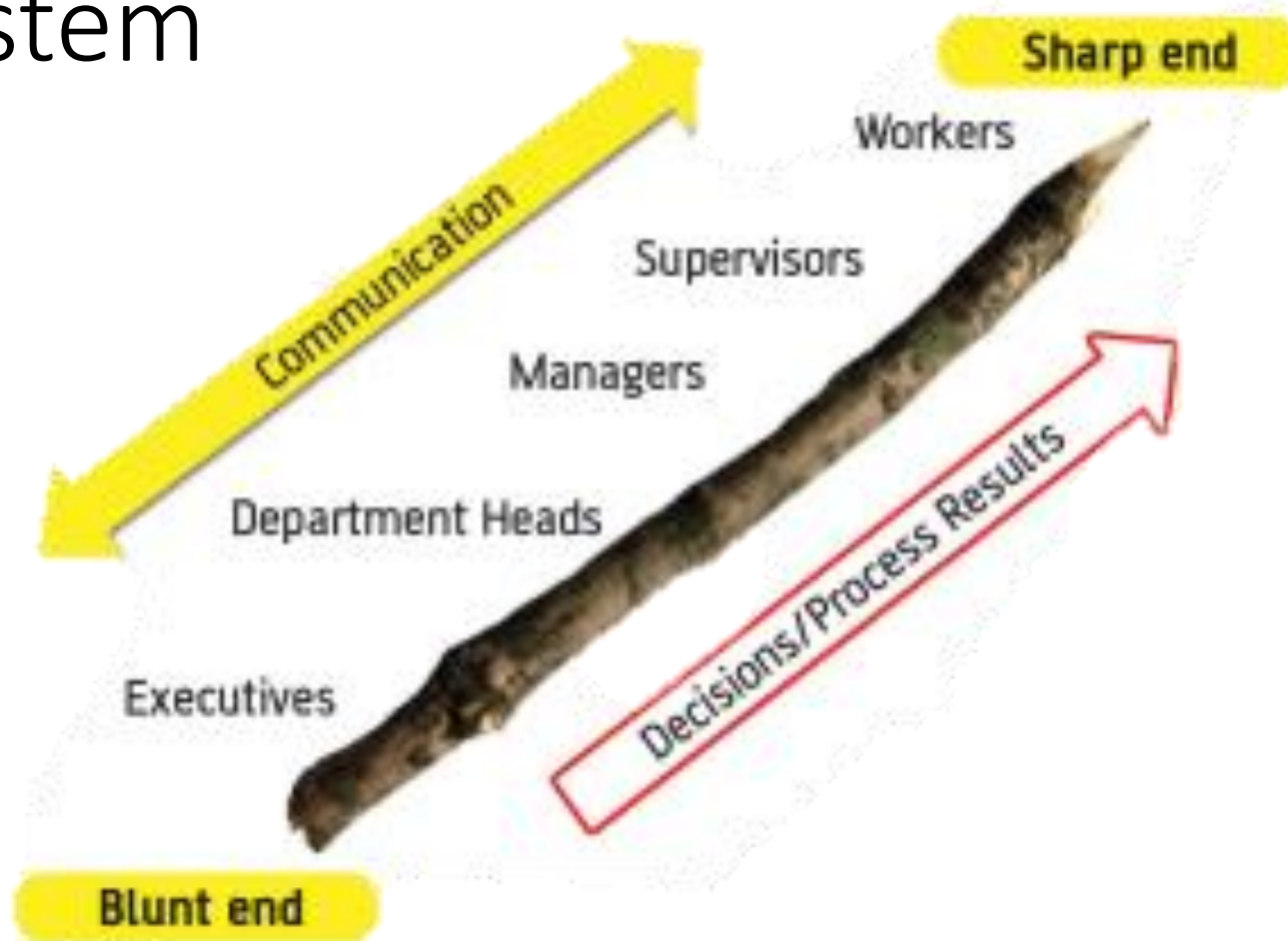


Many Errors Fall into Recurrent Patterns

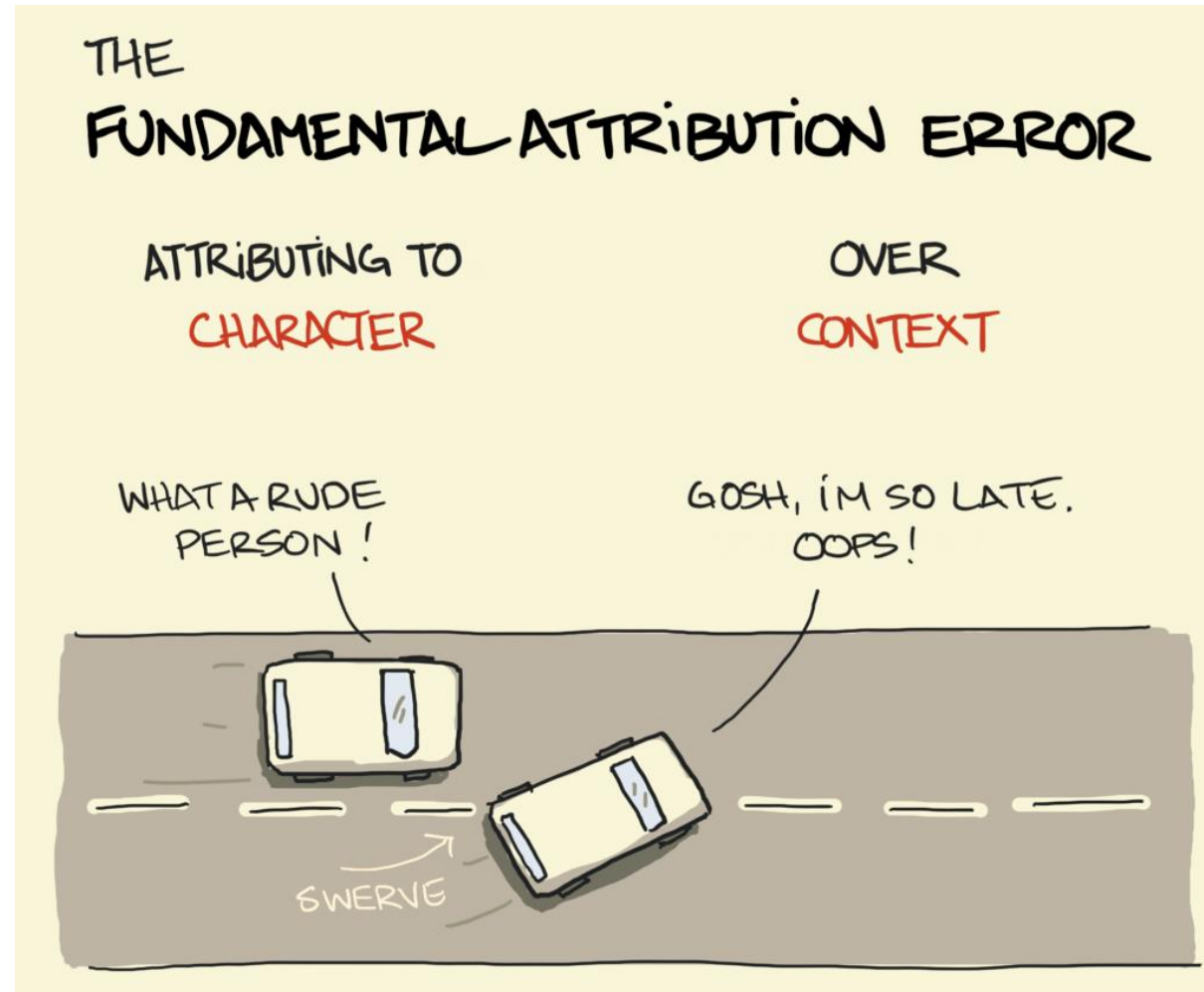


Reassembly & Installation associated with large majority of errors made

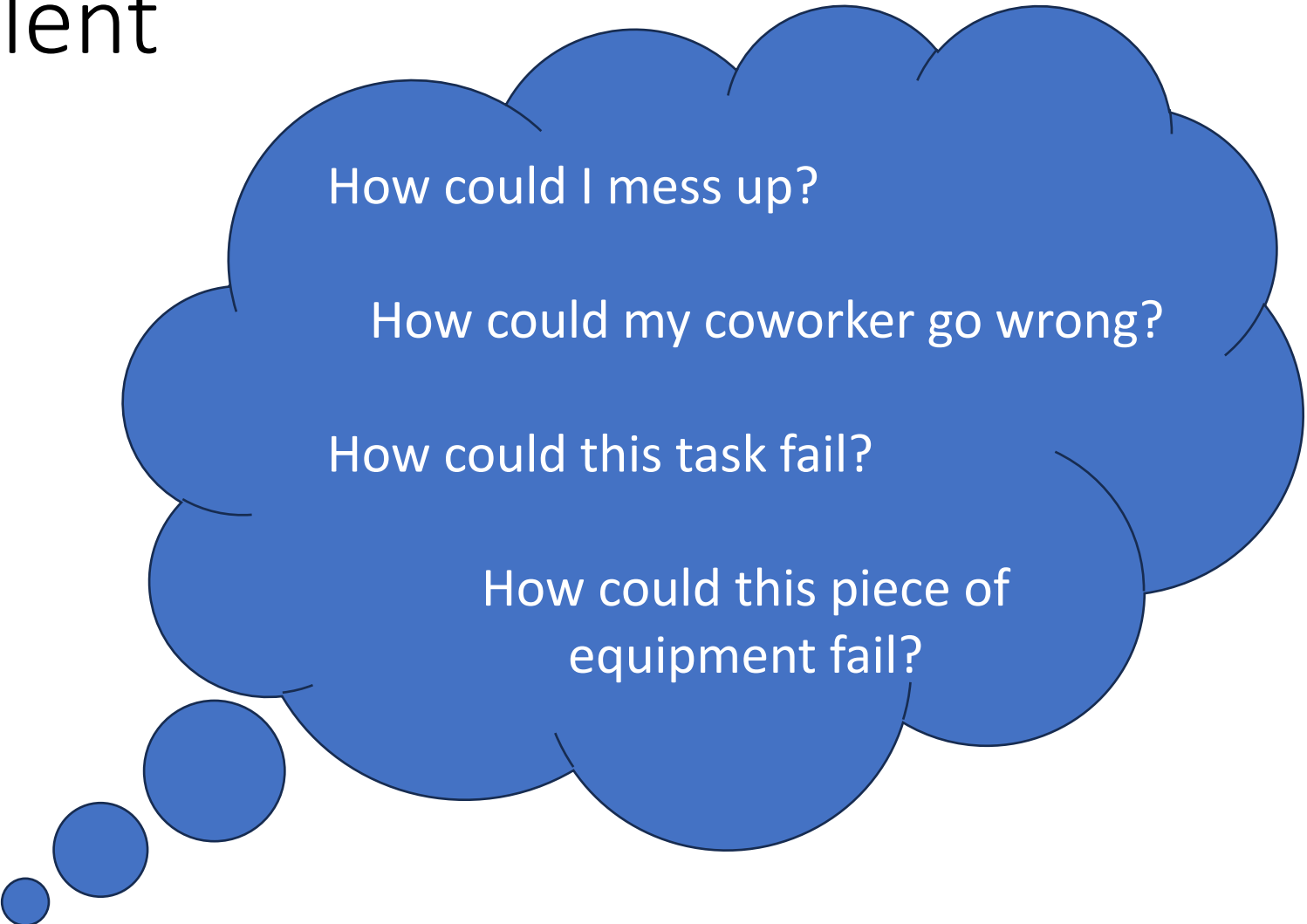
Safety-significant Errors can occur at all Levels of the System



Error Management is about Managing the Manageable



Error Management is about Making Good People Excellent



How could I mess up?

How could my coworker go wrong?

How could this task fail?

How could this piece of equipment fail?

Effective Error Management aims at Continuous Reform rather than Local Fixes

Old View

What is safety?

The absence of injuries and illnesses.

Safety would be improved if management and workers were just more committed.

Our perspective of workers

We need to help workers be safe.

Accident investigation

Who did it?

Find an unsafe act of a person.

Find the root cause.

What we look for

Looking for hazards is all we need to do.

New View

What is safety?

Safety is an emergent system property, which is improved by improving the system.

Management and workers are both affected by the intense economic pressure, which constrains resources and leads to continual process changes that can often impact safety in subtle ways.

Our perspective of workers

Workers need to help us understand how work is done and how the system affects it.

Workplace learning

How did this happen?

Discover the context.

Use a systems approach to find the multiple contributing safety and operational factors.

Analyze and solve

What we look for

Looking for hazards is important but not enough.

Processes must be in place that make us "masters of the blue line". Understanding how work is actually performed provides critical insight to organizational weaknesses and deficiencies that undermine safety and operational performance.

There is no One Best Way

Human Factors:

- Human factors is the study of how humans interact with systems and equipment, and how to design systems to be more user-friendly and safe. 
- It is a key area of focus for both HOP and SMS, as human performance is a critical factor in safety and overall system effectiveness. 

Relationship between HOP and SMS:

- HOP principles can be used to improve the effectiveness of SMS by focusing on human performance factors. 
- SMS provides the framework for managing safety risks, while HOP helps to understand and address the human factors that contribute to those risks. 
- By integrating HOP principles into SMS, organizations can create a more robust and effective safety management system. 